

The existence theorem
SGA 3, Exposé XXV

English draft

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About this draft

This is an unofficial English translation of Exposé XXV, “The existence theorem” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF Exp25-13oct24.pdf (11 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % `typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % `typo?:`. Scholarly proofreading remains outstanding.

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Exposé XXV. The existence theorem

⁽⁰⁾ For completeness, we give in this exposé a proof of the existence theorem for split groups. Like Chevalley’s original proof (“Sur certains schémas de groupes semi-simples”, Séminaire Bourbaki, May 1961, no. 219), it rests on the existence of complex semisimple algebraic groups of all possible types. The principle of a more satisfactory proof, proving directly the existence of a simply connected split semisimple $\mathbf{Z}$ -group corresponding to a given Cartan matrix, was given by Cartier (unpublished).⁽¹⁾ We point out, however, that the difficulty is not to give an explicit construction of a group scheme, but to verify that the group thus constructed does satisfy the required conditions, that is, essentially, that its fibers are indeed smooth and reductive.

1. Statement of the theorem

Theorem 1.1. *Let S be a nonempty scheme. The functor*

$$G \longmapsto \mathcal{R}(G)$$

is an equivalence from the category of pinned reductive S -groups to the category of pinned reduced root data.

By the uniqueness theorem (Exp. XXIII, 4.1), the preceding statement is equivalent to:

Corollary 1.2 (Existence of split groups). *For every reduced root datum $\mathcal{R}$, there exists a split reductive $\mathbf{Z}$ -group G such that $\mathcal{R}(G) \simeq \mathcal{R}$.⁽²⁾*

In particular:

Corollary 1.3. *Let k be a field. For every splittable reductive k -group G_k , there exists a reductive $\mathbf{Z}$ -group G such that $G \otimes_{\mathbf{Z}} k \simeq G_k$.*

Conversely, let us first note that, to prove 1.2, it suffices by Exp. XXII 4.3.1 and Exp. XXI 6.5.10 to consider the case where the root datum $\mathcal{R}$ is simply connected (and even irreducible, if one wishes, by Exp. XXI, 7.1.6).

Moreover, under the conditions of 1.3, the group scheme G has constant type (since $\mathrm{Spec}(\mathbf{Z})$ is connected), hence type $\mathcal{R}(G_k)$; by Exp. XXIII, 5.9, it follows that the validity of 1.3 for a given group G_k implies the existence of a split $\mathbf{Z}$ -group of type $\mathcal{R}(G_k)$.⁽³⁾

⁽⁰⁾**N.D.E.** Version of 13/10/2024.

⁽¹⁾**N.D.E.** Such a principle was also outlined by B. Kostant [Ko66]; a complete proof, using quantum groups, was given recently by G. Lusztig [Lu09].

⁽²⁾**N.D.E.** We also point out that the work [BT84] of F. Bruhat and J. Tits contains a variant of Chevalley’s construction (*loc. cit.*, 2.2.3–2.2.5 and § 3.2), which provides in particular a smooth affine $\mathbf{Z}$ -group G with connected fibers, possessing a split maximal torus, and whose generic fiber is a reductive $\mathbf{Q}$ -group of type $\mathcal{R}$; the fact that the geometric fibers of G are reductive follows from the study of the unipotent radical of a special fiber made in *loc. cit.*, 4.6.12 and 4.6.15 (valid for more general G , associated with a valued root datum), but it is simpler to deduce it from the description of $\mathfrak{g} = \mathrm{Lie}(G)$, from Exp. XIX 1.12 (iii), and from the existence of the elements $w_\alpha(X)$ of Exp. XX 3.1 (iv) (cf. [BT84], 3.2.1).

⁽³⁾**N.D.E.** In fact, Exp. XXIII, 5.9 is not necessary, since the present exposé constructs, for every semisimple $\mathbf{Q}$ -group $G_{\mathbf{Q}}$, a semisimple $\mathbf{Z}$ -group G of the same type as $G_{\mathbf{Q}}$, equipped with a split maximal torus T ; thus, by Exp. XXII 2.2, G is split.

To prove 1.2 and hence 1.1, it therefore suffices to prove 1.3 when k has characteristic zero (for example $k = \mathbf{C}$) and G_k is simply connected (and in particular semisimple), as well as:

Proposition 1.4. *For every simply connected reduced root datum $\mathcal{R}$, there exists a semisimple algebraic $\mathbf{C}$ -group of type $\mathcal{R}$.*

One can prove 1.4 as follows. First, one knows that there exists a complex semisimple Lie algebra $\mathfrak{g}$ of type $\mathcal{R}$, cf. for example N. Jacobson, *Lie Algebras*, ch. VII, Th. 5.⁽⁴⁾ Then $G = \text{Aut}(\mathfrak{g})^0$ is a semisimple algebraic $\mathbf{C}$ -group of type $\text{ad}(\mathcal{R})$.⁽⁵⁾ By Bible, § 23.1, prop. 1, one deduces the existence of a semisimple $\mathbf{C}$ -group of type $\mathcal{R}$.

The remainder of this exposé is devoted to proving 1.3 for k of characteristic zero and G_k semisimple. This will be done in two stages: construction of a “chunk of a group $\mathbf{Z}$ -scheme” (no. 2), study of the group obtained by applying the “Weil theorem” (no. 3).

To avoid confusion, *we shall not use in no. 2* the abbreviated notation X_k to denote the k -scheme $X \otimes_{\mathbf{Z}} k$, where X is a $\mathbf{Z}$ -scheme.

2. Existence theorem: construction of a group chunk

2.1. — Let us choose once and for all a splitting of G_k , denoted

$$(G_k, T_k, M, R)$$

(cf. Exp. XXII, 1.13), a system of simple roots Δ of R (defining the system of positive roots R_+), and a Chevalley system $(X_{\alpha,k})_{\alpha \in R}$ of G_k (Exp. XXIII, 6.1 and 6.2) satisfying the following additional condition (cf. XX 2.6): for every $\alpha \in R$, one has

$$X_{\alpha,k} X_{-\alpha,k} = 1.$$

Finally, let us choose on the subgroup of M generated by R a total ordering compatible with the group structure, such that the roots > 0 are the elements of R_+ . The roots are then denoted

$$-\alpha_n < -\alpha_{n-1} < \cdots < -\alpha_1 < \alpha_1 < \alpha_2 < \cdots < \alpha_n.$$

For $\alpha \in R$, one denotes by $U_{\alpha,k}$ the vector group corresponding to the root α , and by

$$p_{\alpha,k} : \mathbf{G}_{a,k} \xrightarrow{\sim} U_{\alpha,k}$$

the isomorphism of vector groups defined by $X_{\alpha,k}$.

2.2. — The splitting of G_k includes in particular an isomorphism of k -groups

$$T_k \simeq D_k(M).$$

Put $T = D(M)$; this is a $\mathbf{Z}$ -torus, and one can regard the preceding isomorphism as an isomorphism

$$T_k \simeq T \otimes_{\mathbf{Z}} k.$$

⁽⁴⁾**N.D.E.** Let Δ be a base of R , and let $\tilde{\mathfrak{g}}$ be the $\mathbf{C}$ -Lie algebra generated by generators $(e_\alpha, f_\alpha, h_\alpha)_{\alpha \in \Delta}$ subject to the relations $[h_\alpha, h_\beta] = 0$, $[e_\alpha, f_\beta] = h_\alpha$ if $\beta = \alpha$ and $= 0$ otherwise, $[h_\alpha, e_\beta] = (\alpha^*, \beta)e_\beta$ and $[h_\alpha, f_\beta] = -(\alpha^*, \beta)f_\beta$. In *loc. cit.*, $\mathfrak{g}$ is defined as the quotient of $\tilde{\mathfrak{g}}$ by the intersection of the kernels of the finite-dimensional irreducible representations of $\tilde{\mathfrak{g}}$; in [Se66], § VI.5, Th. 9 (see also [BLie], VIII, § 4.3, Th. 1), it is shown that $\mathfrak{g}$ is the quotient of $\tilde{\mathfrak{g}}$ by the relations $\text{ad}(e_\alpha)^{1-(\alpha^*, \beta)}(e_\beta) = 0$ and $\text{ad}(f_\alpha)^{1-(\alpha^*, \beta)}(f_\beta) = 0$. For an explicit description of the structure constants (in particular the choice of signs), see Exp. XXIII 6.5 and 6.7, as well as [Ti66], § 4, Th. 1, and (for types A,D,E) [Sp98], 10.2.5.

⁽⁵⁾**N.D.E.** One can suppose $\mathcal{R}$ irreducible, hence $\mathfrak{g}$ simple. By a general argument, one knows that $\text{Lie}(G)$ is the $\mathbf{C}$ -Lie algebra of derivations of $\mathfrak{g}$ (cf. [DG70], § II.4, Prop. 2.3); these are all inner (cf. [BLie], I § 6.1, cor. 3 of prop. 1), so $\text{Lie}(G) = \mathfrak{g}$; consequently G has no invariant subgroup of dimension > 0 , hence G is semisimple. Its root system is then the same as that of $\mathfrak{g}$; moreover the center of G acts both trivially and faithfully on $\mathfrak{g}$, and so is trivial, hence G is adjoint.

One has

$$\mathrm{Hom}_{\mathbf{Z}\text{-gr.}}(T, \mathbf{G}_m) = M,$$

and one will regard the elements of $R \subset M$ as characters of T . Similarly, one will regard the elements of R^* as morphisms of $\mathbf{Z}$ -groups $\mathbf{G}_m \rightarrow T$.

2.3. — For each $\alpha \in R_+$, let $\mathbf{G}_a(\alpha)$ be a copy of the group $\mathbf{G}_a$; consider the $\mathbf{Z}$ -scheme

$$U = \mathbf{G}_a(\alpha_1) \times \cdots \times \mathbf{G}_a(\alpha_n).$$

If U_k denotes the unipotent part of the Borel group B_k of G_k defined by R_+ , denote by

$$a : U \otimes_{\mathbf{Z}} k \xrightarrow{\sim} U_k$$

the isomorphism of k -schemes defined by

$$a(x_1, \dots, x_n) = p_{\alpha_1, k}(x_1) \cdots p_{\alpha_n, k}(x_n).$$

2.4. — The group law of U_k is expressed by relations of the form

$$a(x_1, \dots, x_n) \cdot a(y_1, \dots, y_n) = a(z_1, \dots, z_n),$$

where each z_h ($h = 1, \dots, n$) is expressed as a polynomial

$$z_h = x_h + y_h + Q_h(x_1, \dots, x_{h-1}, y_1, \dots, y_{h-1}),$$

the coefficients of Q_h being integers (Exp. XXII, 5.5.8 and Exp. XXIII, 6.4). Moreover $Q_h(x_1, \dots, x_{h-1}, 0, \dots, 0) = 0$.

Let us equip U with the composition law defined by the preceding formulas (which are indeed “defined over $\mathbf{Z}$ ”). Since this law induces on U_k its group law, it is associative, and (0) is an identity element for it (indeed, the preceding two assertions are expressed by relations among the polynomials Q_h , and $\mathbf{Z} \rightarrow k$ is injective). Let us show that it is a group law: if (x_i) is a section of U (over any S), one computes the inverse (y_i) of (x_i) by the recurrence formulas:

$$y_i = -x_i - Q_i(x_i, \dots, x_{i-1}, y_1, \dots, y_{i-1})$$

which are again “defined over $\mathbf{Z}$ ”.

To sum up, we have constructed on U a group law such that the preceding isomorphism a is an isomorphism of groups.

For each $\alpha \in R_+$, consider the morphism

$$p_\alpha : \mathbf{G}_a \rightarrow U$$

defined by $p_\alpha(x) = (x_i)$, where

$$\begin{cases} x_i = x & \text{if } \alpha_i = \alpha, \\ x_i = 0 & \text{if } \alpha_i \neq \alpha. \end{cases}$$

This is a closed immersion and a group homomorphism; its image is denoted U_α . One has $(x_i) = p_{\alpha_1}(x_1) \cdots p_{\alpha_n}(x_n)$, which proves that U is identified with the product

$$U = U_{\alpha_1} \cdot U_{\alpha_2} \cdots U_{\alpha_n}.$$

2.5. — Let us make $T = D(M)$ act on each U_α through the character α ; one immediately verifies that this defines an action of T on the group U , and one can construct the semidirect product $B = T \cdot U$. One has a canonical isomorphism of k -groups

$$B \otimes_{\mathbf{Z}} k \xrightarrow{\sim} B_k.$$

If we now take any ordering on R_+ , the morphism

$$\prod_{\alpha \in R_+} U_\alpha \rightarrow U$$

defined by the product in U is still an isomorphism. Indeed, since both sides are flat $\mathbf{Z}$ -schemes of finite presentation, one can content oneself with verifying the assertion on the geometric fibers; one is then reduced to Lazard’s theory (Bible, § 13.1, prop. 1): one regards U as a

group with operators T , and uses the fact that the U_α are pairwise nonisomorphic as groups with operators (since the characters $\alpha \in R_+$ of T are pairwise distinct on each fiber).

2.6. — Replacing R_+ by $R_- = -R_+$, one similarly constructs groups U^- , B^- , U_α ($\alpha \in R_-$), and isomorphisms

$$p_\alpha : \mathbf{G}_a \xrightarrow{\sim} U_\alpha, \quad \alpha \in R_-.$$

Finally, introduce the product scheme

$$\Omega = U^- \times T \times U;$$

one has a canonical isomorphism of k -schemes

$$\Omega \otimes_{\mathbf{Z}} k \xrightarrow{\sim} U_k^- \times_k T_k \times_k U_k \simeq \Omega_k,$$

where Ω_k is the “big cell” of G_k (Exp. XXII, 4.1.2).

From now on, we identify $\Omega \otimes_{\mathbf{Z}} k$ with Ω_k by the preceding isomorphism; we regard U^- , T , U as subschemes of Ω , by means of the identity sections. One denotes by $\underline{e} = ((0), e, (0))$ the “identity section” of Ω .

Our aim now is to put a group-chunk law on Ω .

Lemma 2.7. *Let $\alpha \in \Delta$, and let $w_{\alpha,k}$ be the element of $\text{Norm}_{G_k}(T_k)(k)$ defined by $X_{\alpha,k}$ (recall that by definition one has*

$$w_{\alpha,k} = p_{-\alpha,k}(-1)p_{\alpha,k}(1)p_{-\alpha,k}(-1)).$$

There exists an open subset V_α of Ω , containing the section $\underline{e}$, and a morphism

$$h_\alpha : V_\alpha \longrightarrow \Omega,$$

satisfying the following conditions:

- (i) $h_\alpha(\underline{e}) = \underline{e}$,
- (ii) $(h_\alpha) \otimes_{\mathbf{Z}} k$ coincides with the restriction of $\text{int}(w_{\alpha,k})$ to $V_\alpha \otimes_{\mathbf{Z}} k \subset G_k$.
- (iii) One has $T \subset V_\alpha$, and h_α sends T into T . For every $\beta \in R$, one has $U_\beta \subset V_\alpha$, and h_α sends U_β into $U_{s_\alpha(\beta)}$.

By the definition of a Chevalley system (Exp. XXIII, 6.1), there exists for each $\beta \in R$ an integer $e_\beta = \pm 1$ such that

$$\text{int}(w_{\alpha,k})p_{\beta,k}(x) = p_{s_\alpha(\beta),k}(e_\beta x)$$

for every $x \in \mathbf{G}_a(S)$, $S \longrightarrow \text{Spec}(k)$.

Let S be any scheme; let us write any element of $\Omega(S)$ in the form

$$u = \left(\left(\prod_{\substack{\beta \in R_- \\ \beta \neq -\alpha}} p_\beta(x_\beta) \right) \cdot p_{-\alpha}(x_{-\alpha}), \quad t, \right. \\ \left. p_\alpha(x_\alpha) \cdot \left(\prod_{\substack{\beta \in R_+ \\ \beta \neq \alpha}} p_\beta(x_\beta) \right) \right),$$

where one has chosen an ordering (any ordering) on $R_- - \{-\alpha\}$ and $R_+ - \{\alpha\}$ (cf. 2.5).

One defines a morphism $d : \Omega \longrightarrow \text{Spec}(\mathbf{Z})$ by

$$d(u) = \alpha(t) + x_\alpha x_{-\alpha}.$$

Let V_α be the open subset Ω_d (that is, the open subset of Ω defined by “ $d(u)$ invertible”); it contains $\underline{e}$, T , and each U_β , $\beta \in R$. Let

$$h_\alpha : V_\alpha \longrightarrow \Omega$$

be the morphism defined by $h_\alpha(u) = (a(u), b(u), c(u))$, where

$$\begin{aligned} a(u) &= \left(\prod_{\substack{\beta \in R_- \\ \beta \neq -\alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right) \cdot p_{-\alpha}(-x_\alpha d(u)^{-1}), \\ b(u) &= t \cdot \alpha^*(d(u)), \\ c(u) &= p_\alpha(-x_{-\alpha} d(u)^{-1}) \cdot \left(\prod_{\substack{\beta \in R_+ \\ \beta \neq \alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right). \end{aligned}$$

Since s_α permutes the positive (resp. negative) roots distinct from α (resp. $-\alpha$), $c(u)$ (resp. $a(u)$) is then a section of U (resp. U^-), and the preceding morphism is well defined. It trivially satisfies (i) and (iii). As for (ii), this follows immediately from the definition of the e_β , $\beta \in R$, and from Exp. XX, 3.12.

Lemma 2.8. *There exist open subsets V and V' of Ω , and morphisms*

$$h : V \longrightarrow \Omega, \quad h' : V' \longrightarrow \Omega,$$

satisfying the following conditions:

- (i) V and V' contain $\underline{e}$, and $h(\underline{e}) = h'(\underline{e}) = \underline{e}$.
- (ii) The morphism induced by $h' \circ h : h^{-1}(V') \longrightarrow \Omega$ is the restriction of the identity morphism.
- (iii) The k -morphisms $h \otimes_{\mathbf{Z}} k$ and $h' \otimes_{\mathbf{Z}} k$ are the restrictions to $V \otimes_{\mathbf{Z}} k$ and $V' \otimes_{\mathbf{Z}} k$ of automorphisms of the group G_k .
- (iv) V and V' contain U , T and U^- ; h and h' send U into U^- , U^- into U , and T into T .

Let $\bar{w}_0$ be the element of the Weyl group of G_k which transforms R_+ into R_- . Write

$$\bar{w}_0 = s_{\alpha_n} \cdots s_{\alpha_1}, \quad \alpha_i \in \Delta$$

(no connection with the numbering of 2.1). Put

$$w_0 = w_{\alpha_n, k} \cdots w_{\alpha_1, k} \in \underline{\text{Norm}}_{G_k}(T_k)(k).$$

Define by induction on $i \leq n$ an open subset V_i of Ω and a morphism $h_i : V_i \longrightarrow \Omega$ by $V_0 = \Omega$, $h_0 = \text{id}$, and, for $i = 0, \dots, n-1$,

$$V_{i+1} = h_i^{-1}(V_{\alpha_{i+1}}), \quad h_{i+1} = h_{\alpha_{i+1}} \circ h_i,$$

where the notations V_{α_j} and h_{α_j} are those of 2.7.

Take $V = V_n$ and $h = h_n$. The conditions of (i), (iii) and (iv) concerning V and h are indeed satisfied; for (i) and (iii) this follows immediately from 2.8, and for (iv), from the fact that $h \otimes_{\mathbf{Z}} k$ is the restriction of $\text{int}(w_0)$ to $V \otimes_{\mathbf{Z}} k$.

Since $(\bar{w}_0)^2 = 1$, one also has

$$\bar{w}_0 = s_{\alpha_1} \cdots s_{\alpha_n} = (s_{\alpha_1}^3) \cdots (s_{\alpha_n}^3).$$

Putting

$$w'_0 = (w_{\alpha_1, k})^3 \cdots (w_{\alpha_n, k})^3,$$

and carrying out the same construction as above, one deduces an open subset V' and a morphism h' likewise satisfying (i), (iii), (iv). Moreover, $h' \otimes_{\mathbf{Z}} k$ is the restriction of $\text{int}(w'_0)$ to $V' \otimes_{\mathbf{Z}} k$. But for each simple root $\alpha \in \Delta$, one has $(w_{\alpha, k})^4 = e$ (cf. Exp. XX, 3.1), so $w'_0 \cdot w_0 = e$, which shows that $h' \circ h$ induces the identity morphism in a nonempty open subset of $\Omega \otimes_{\mathbf{Z}} k$. But since Ω is smooth and of finite presentation over $\mathbf{Z}$, $\Omega \otimes_{\mathbf{Z}} k$ is schematically dense in Ω , which proves (ii).

Proposition 2.9. *There exist an open subset V_1 of $\Omega \times \Omega$, an open subset V_2 of Ω , and morphisms*

$$\pi : V_1 \longrightarrow \Omega, \quad \sigma : V_2 \longrightarrow \Omega,$$

possessing the following properties:

- (i) If $x \in \Omega(S)$, then $(\underline{e}, x)$ and $(x, \underline{e})$ are sections of V_1 , and

$$\pi(\underline{e}, x) = \pi(x, \underline{e}) = x.$$

- (ii) V_2 contains $\underline{e}$, and $\sigma(\underline{e}) = \underline{e}$.

- (iii) π_k and σ_k are the restrictions of the morphisms $G_k \times_k G_k \rightarrow G_k$ and $G_k \rightarrow G_k$ defined by the product (resp. the inverse).

PROOF. Let (v, t, u) and (v', t', u') be two sections of Ω . Then $h(u)$ is a section of U^- , $h(v')$ is a section of U by 2.8 (iv), and one can therefore regard $(h(u), e, h(v'))$ as a section of Ω . Let V_1 be the open subset of $\Omega \times \Omega$ defined by the condition:

$$((v, t, u), (v', t', u')) \in V_1(S) \iff (h(u), e, h(v')) \in V'(S)$$

(notations of 2.8). If $((v, t, u), (v', t', u'))$ is a section of V_1 , then $h'(h(u), e, h(v'))$ is defined; this is a section of Ω which one can decompose:

$$h'(h(u), e, h(v')) = (v'', t'', u'').$$

One then puts

$$\pi((v, t, u), (v', t', u')) = (v \cdot tv''t^{-1}, tt''t', t'^{-1}u''t' \cdot u').$$

The verification of (i) is immediate (by 2.8 (ii)). To verify the condition of (iii) concerning π , one sees that

$$h'(h(u), e, h(v')) = uv' = v''t''u''$$

when $u \in U(S)$, $v \in U^-(S)$, $S \rightarrow k$, by 2.8 (iii) and (ii). One constructs σ in a similar manner: if (v, t, u) is a section of Ω , $h(u^{-1})$ is a section of U^- , $h(v^{-1})$ a section of U , $h(t^{-1})$ a section of T , so $(h(u^{-1}), h(t^{-1}), h(v^{-1}))$ is a section of Ω , and one can define an open subset V_2 of Ω by

$$(v, t, u) \in V_2(S) \iff (h(u^{-1}), h(t^{-1}), h(v^{-1})) \in V'(S)$$

and a morphism $\sigma : V_2 \rightarrow \Omega$ by

$$\sigma(v, t, u) = h'(h(u^{-1}), h(t^{-1}), h(v^{-1})).$$

One verifies the conditions on σ as above. $\square$

Corollary 2.10. π is “generically associative”, and σ is a “generic inverse”: if $x, y, z \in \Omega(S)$ and if the expressions below are defined (which always occurs above an open subset of Ω containing the identity section), one has:

$$\pi(x, \pi(y, z)) = \pi(\pi(x, y), z), \quad \pi(x, \sigma(x)) = \underline{e} = \pi(\sigma(x), x).$$

Indeed, the two sides of each of these formulas define morphisms between smooth $\mathbf{Z}$ -schemes of finite presentation, which coincide on the generic fibers, by 2.10 (iii).

Corollary 2.11. Let $\alpha \in R$. For every S and all $x, y \in \mathbf{G}_a(S)$ such that $(p_\alpha(x), p_{-\alpha}(y)) \in V_1(S)$ and $1 + xy \in \mathbf{G}_m(S)$ (which defines an open subset of $\mathbf{G}_{a,S}^2$ containing the section $(0, 0)$), one has:

$$\pi(p_\alpha(x), p_{-\alpha}(y)) = \left(p_{-\alpha}\left(\frac{y}{1+xy}\right), \alpha^*(1+xy), p_\alpha\left(\frac{x}{1+xy}\right) \right).$$

The proof is the same as before, by Exp. XX, 2.1.

3. Existence theorem: end of the proof

To simplify the language, let us make the following definition.

Definition 3.1. Let S be a scheme and G a group S -scheme. One says that G is *admissible* if there exists an open immersion of S -schemes $i : \Omega_S = \Omega \times S \rightarrow G$ satisfying the following conditions:

(i) The diagram

$$\begin{array}{ccc} (V_1)_S & \longrightarrow & \Omega_S \times_S \Omega_S \xrightarrow{i \times_S i} G \times_S G \\ \pi \downarrow & & \downarrow \pi_G \\ \Omega_S & \xrightarrow{i_S} & G, \end{array}$$

where π_G denotes the multiplication morphism in G , is commutative.

(ii) There exists a finite set of sections $a_j \in \Omega(S)$ such that the $i(a_j) \cdot i(\Omega_S)$ cover G .

By the “Weil theorem” (Exp. XVIII, 3.13 (iii) and (iv)), one has:

Lemma 3.2. *If, for every scheme S étale and of finite type over $\mathbf{Z}$, every admissible S -group is affine, then there exists an admissible affine $\mathbf{Z}$ -group.*

Now one has:

Lemma 3.3. *Let S be a scheme and G an admissible S -group. Then G is smooth and of finite presentation over S , with affine connected semisimple fibers.*

Since Ω_S is smooth and of finite presentation over S , with connected fibers, the same holds for G , by condition (ii). To verify 3.3, one can therefore suppose that S is the spectrum of a field K . Identify Ω_K with its image in G . It is clear that Ω_K is the product

$$\prod_{\alpha \in R_-} U_{\alpha,K} \cdot T_K \cdot \prod_{\alpha \in R_+} U_{\alpha,K}$$

of the subgroups T_K and $U_{\alpha,K}$ ($\alpha \in R$) of G . The Lie algebra of G is therefore identified with the direct sum

$$\mathrm{Lie}(T_K) \oplus \coprod_{\alpha \in R} \mathrm{Lie}(U_{\alpha,K}).$$

Since the inner automorphism defined by a section of T_K acts on $U_{\alpha,K}$, and hence on $\mathrm{Lie}(U_{\alpha,K})$, through the character

$$\alpha \in R \subset M \simeq \mathrm{Hom}_{K\text{-gr.}}(T_K, \mathbf{G}_{m,K}),$$

the preceding decomposition of $\mathrm{Lie}(G)$ is exactly the decomposition under the adjoint action of T . The roots of G_K relative to T_K are therefore the $\alpha \in R$. Apply Exp. XIX, 1.13. Let T_α be the maximal torus of $\mathrm{Ker}(\alpha) \subset T_K$, and let $Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha)$; it suffices for us to prove that each Z_α is reductive. Now $Z_\alpha \cap \Omega_K$ is precisely

$$\prod_{\substack{\beta \in R_- \\ \beta|T_\alpha=e}} U_{\beta,K} \cdot T_K \cdot \prod_{\substack{\beta \in R_+ \\ \beta|T_\alpha=e}} U_{\beta,K};$$

but the roots vanishing on T_α are the rational multiples of α , hence α and $-\alpha$; this proves

$$Z_\alpha \cap \Omega_K = U_{-\alpha,K} \cdot T_K \cdot U_{\alpha,K}.$$

To prove that Z_α is reductive, it suffices, by Exp. XX 3.4, to prove that $U_{\alpha,K}$ and $U_{-\alpha,K}$ do not commute, which follows immediately from 2.11.

It follows from 3.2 and 3.3 that the proof will be complete if we prove:

Lemma 3.4. *If S is a locally noetherian scheme of dimension ≤ 1 , and if G is a smooth S -group of finite type, with affine connected semisimple fibers, then G is affine (and hence semisimple).*

Nota. — In Exp. XVI, one saw that 3.4 holds without any hypothesis on S , but the proof is relatively delicate; since here we need only the special case 3.4, we give a direct proof of it.

Consider the Lie algebra $\mathfrak{g}$ of G , which is a locally free $\mathcal{O}_S$ -module, and the adjoint representation of G

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}_{\mathcal{O}_S}(\mathfrak{g}).$$

To prove that G is affine over S , it suffices to prove that the morphism Ad is affine. Since G is smooth with connected fibers, it is separated over S (VIB 5.5), hence the morphism Ad is separated. Using a result proved in the appendix (see 4.1), it suffices to prove that the morphism Ad is quasi-finite. One is therefore reduced to the case where S is the spectrum of a field; in this case G is affine, hence semisimple, and one is reduced to Exp. XXII 5.7.14.

4. Appendix

We used the following proposition in the course of the proof:

Proposition 4.1. *Let S be a locally noetherian scheme of dimension ≤ 1 , G and H two group S -schemes of finite type, and $f : G \rightarrow H$ a quasi-finite separated group morphism. If G is flat over S ,⁽⁶⁾ then f is affine.*

We shall give the proof only in the case where G is smooth over S , a hypothesis which is indeed satisfied in the application we have made of the proposition.

4.2. — By EGA II, 1.6.4, one can suppose S reduced. By the usual techniques of passage to the limit,⁽⁷⁾ one can suppose S local. If $\dim(S) = 0$, the assertion is trivial;⁽⁸⁾ suppose $\dim(S) = 1$. By faithfully flat descent, one can suppose that S is complete with algebraically closed residue field. Replacing S by its normalization $\tilde{S}$, one can (EGA II, 6.7.1 and EGA 0_{IV} 23.1.5) suppose S normal.⁽⁹⁾ One is therefore reduced to the case where S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field.

4.3. — Let η (resp. s) be the generic (resp. closed) point of S . Consider the image $f_\eta(G_\eta)$ of G_η in H_η . This is a closed group subscheme of H_η . Let H' be the schematic closure in H of $f_\eta(G_\eta)$. Since $H' \rightarrow H$ is affine (it is a closed immersion), one can replace H by H' , and hence suppose H flat over S and f_η surjective. Since f_η is finite, and G and H flat over S , one has

$$\dim(G_s) = \dim(G_\eta) = \dim(H_\eta) = \dim(H_s).$$

4.4. — Let $H_s^0, \dots, H_s^n$ be the irreducible components of H_s , where H_s^0 denotes the identity component, and let $z_0, \dots, z_n$ be their generic points. Since each local ring $\mathcal{O}_{H, z_i}$ has dimension ≤ 1 , the morphism $G \times_H \mathcal{O}_{H, z_i} \rightarrow \mathcal{O}_{H, z_i}$ is affine, since it is quasi-finite and separated (cf. Exp. XVI, lemma 4.2), hence G is affine over H in a neighborhood of z_i . Denoting by V the largest open subset of H such that $G|_V$ is affine over V , it follows that V contains all the z_i ,⁽¹⁰⁾ hence contains at least one closed point $y_i \in H_s^i$ (and one has $\kappa(y_i) = \kappa(s)$, since $\kappa(s)$ is algebraically closed).

On the other hand, V is evidently stable under the translation defined by any element $g \in G(S)$. But one has $\dim(G_s) = \dim(H_s)$ and f_s is finite, so

$$f(s) : G_s^0(s) \rightarrow H_s^0(s)$$

⁽⁶⁾**N.D.E.** We have added the flatness hypothesis, which had been omitted.

⁽⁷⁾**N.D.E.** Cf. EGA IV₃, 8.10.5 (viii).

⁽⁸⁾**N.D.E.** Indeed, if $S = \text{Spec}(k)$ (k a field), then f is the composite of the projection $p : G \rightarrow G/\text{Ker}(f)$ and a closed immersion i , and since $\text{Ker}(f)$ is finite over k , p is finite (VIB 9.2).

⁽⁹⁾**N.D.E.** Indeed, the irreducible components $S_1, \dots, S_r$ of S are complete integral noetherian local schemes, hence, by a theorem of Nagata (cf. EGA 0_{IV}, 23.1.5), the normalization $\tilde{S}_i$ is finite over S_i , and so $\tilde{S}$ is finite over S . Then, by a theorem of Chevalley (cf. EGA II, 6.7.1), if $f \times_S \tilde{S}$ is an affine morphism, the same holds for f .

⁽¹⁰⁾**N.D.E.** The following phrase has been added.

is surjective. Since A is complete and G smooth over S , the canonical map $G^0(S) \rightarrow G_s^0(s)$ is surjective; since $H_s^0(s)$ acts transitively on each $H_s^i(s)$, it follows that $V \supset H_s(s)$, hence (since $\kappa(s)$ is algebraically closed) $V \supset H_s$.⁽¹¹⁾

Since one evidently has $V \supset H_\eta$, since f_η is finite, one therefore has $V = H$. Q.E.D.

⁽¹¹⁾**N.D.E.** When G is not supposed smooth over S , one can proceed as follows. Let h_0 be a rational point of H_s^0 , and g_0 a rational point of G_s^0 such that $f(g_0) = h_0$. By VIB 5.6.1, there exists a commutative diagram

$$\begin{array}{ccc} S'' & \xrightarrow{g} & G \\ \pi \downarrow & \searrow \phi & \downarrow \\ S' & \xrightarrow{w} & S \end{array}$$

where w is étale and surjective, π finite and surjective, and $\phi^{-1}(s)$ consists of a single point s'' such that $g(s'') = g_0$. Then the morphism $G_{S''} \rightarrow H_{S''}$ is affine above a neighborhood of $h_0 y_i$, and the same holds for $G_{S'} \rightarrow H_{S'}$ (EGA II, 6.7.1), then for $G \rightarrow H$ by faithfully flat descent (EGA IV₂, 2.7.1 (xiii)). Thus $h_0 y_i \in V$, and it follows that V contains $H_s(s)$ and hence H_s .

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⁽¹²⁾**N.D.E.** Additional references cited in this exposé.